

The Epistemic Impossibility of Predictor-Centric Architectures

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December 18, 2025

Abstract

Large language model “hallucinations” are commonly framed as failures introduced by alignment techniques or insufficient training data. In this paper, we argue that this framing is incorrect. We present a formal impossibility result demonstrating that **epistemic honesty is unreachable** for the class of predictor-centric architectures (M_{closed}) aligned via standard reward modeling.

We decompose this failure into two theorems. **Theorem 1 (Representational Impossibility)** proves that a policy conditioning solely on the query cannot satisfy epistemic requirements that depend on external world states. **Theorem 2 (Learnability Impossibility)** proves that even if the architecture is expanded to include retrieval (Grounded Policy), a resource-bounded supervisor cannot distinguish truthful answers from plausible fabrications, rendering the honest policy unlearnable via preference optimization. We validate these theoretical results with forensic artifacts from frontier model interfaces (Grok, Gemini, OLMo), identifying a “**Deadlock of Veracity**” where human operators are architecturally forced to reinforce fabrication.

1 System Definitions

Let $\mathcal{Q}$ be the space of queries, $\mathcal{R}$ be the space of responses (including $\perp$ for abstention), and $\mathcal{W}$ be the set of possible world states.

Definition 1.1 (Observation Model). Let S denote the internal system state, which may include retrieved documents, tool outputs, candidate entities, and intermediate reasoning traces. An observation model is a function $O : S \rightarrow \Omega$ that determines what information is exposed to the supervisor or user.

We say a system operates under a **Text-Only Observation Model** if $O(S) = r \in \mathcal{R}$, i.e., the exported observation consists solely of the linearized response string (optionally including token-level probabilities), and excludes any structured epistemic provenance or causal trace.

Definition 1.2 (Predictor-Centric Policy). For any query $q \in \mathcal{Q}$, $r \in \mathcal{R}$, and world state $w \in \mathcal{W}$, a predictor-centric model is defined by a stochastic policy $\pi : \mathcal{Q} \rightarrow \Delta(\mathcal{R})$, where $\pi(r|q)$ denotes the probability of generating response r given query q . Crucially, π does not take w as an input at inference time.

Definition 1.3 (Truth Status). For any query q and world state $w \in \mathcal{W}$, we define a binary truth status $U(q, w)$:

- $U(q, w) = 1$ (**Determined**): The query has a verifiable answer in w .

- $U(q, w) = 0$ (**Underdetermined**): The query is unanswerable or the answer is unknown in w .

Definition 1.4 (Epistemic Honesty). A policy π is **Epistemically Honest** with margin ϵ relative to world w if:

1. If $U(q, w) = 1$, there exists a correct response r_{corr} such that $\pi(r_{corr}|q) \geq 1 - \epsilon$.
2. If $U(q, w) = 0$, $\pi(\perp|q) \geq 1 - \epsilon$.

2 Theorem 1: Representational Impossibility

This theorem establishes that under a text-only observation model, no policy, regardless of internal architecture, can satisfy Epistemic Honesty across ambiguous world states.

Condition 2.1 (Ambiguity). There exists a query q and two world states $w_A, w_B \in \mathcal{W}$ such that:

1. $U(q, w_A) = 1$ (Answerable in w_A).
2. $U(q, w_B) = 0$ (Unanswerable in w_B).

Theorem 2.1. *For any predictor-centric policy $\pi : \mathcal{Q} \rightarrow \Delta(\mathcal{R})$, it is impossible to satisfy Epistemic Honesty for both world states w_A and w_B simultaneously given the ambiguous query q , provided $\epsilon < 0.5$.*

Proof. 1. The policy $\pi(r|q)$ is a function of q alone. Thus, the distribution over responses is identical in both worlds: $\pi(\cdot|q, w_A) = \pi(\cdot|q, w_B) = \pi(\cdot|q)$.

2. To be honest in w_A , we require $\pi(r_{corr}|q) \geq 1 - \epsilon$.
3. To be honest in w_B , we require $\pi(\perp|q) \geq 1 - \epsilon$.
4. Since $r_{corr} \neq \perp$, these events are disjoint. The sum of probabilities would be $\geq 2(1 - \epsilon)$.
5. If $\epsilon < 0.5$, the sum exceeds 1, which violates the axioms of probability.
6. Therefore, no such policy π exists.

□

Remark: This impossibility holds even if the internal system performs retrieval, tool use, or latent reasoning, so long as the observation model collapses that state to a linearized response.

Implication: A model that cannot “see” the world state (via retrieval or grounding) at inference time cannot be honest about it.

3 Theorem 2: Learnability Impossibility

This theorem establishes that even if the architecture is expanded to allow dependence on w (e.g., via RAG), the training protocol prevents learning the correct dependence if the supervisor is bounded.

Definition 3.1 (Grounded Policy). A grounded policy is a stochastic function $\pi : \mathcal{Q} \times \mathcal{D} \rightarrow \Delta(\mathcal{R})$, where $\mathcal{D}$ represents retrieved documents or observations of w available at inference time. This policy class *can* theoretically represent different behaviors for w_A and w_B .

Definition 3.2 (The Bounded Supervisor). The training process relies on a supervisor S that assigns reward R based on an observation function $Obs(q, r, w)$. The supervisor has a verification budget B .

$$Obs(q, r, w) = \begin{cases} (q, r, \text{Verify}(r, w)) & \text{if } C_{\text{verify}}(q, r, w) \leq B \\ (q, r, \emptyset) & \text{if } C_{\text{verify}}(q, r, w) > B \end{cases}$$

The supervisor’s policy may be human, automated, or hybrid, but is constrained to act solely on the observation $Obs(q, r, w)$ and is subject to a finite verification budget B .

In deployed systems, supervisors—contract annotators, downstream auditors, or automated safety filters—operate under explicit time and cognitive budgets. Public RLHF rater guidelines emphasize relative quality judgments under strict time caps rather than exhaustive fact verification,¹ instantiating the bounded-supervisor assumption in practice. Quantifying the precise relationship between response compositionality and verification time—for example, identifying thresholds beyond which raters resort to sampling—is an important direction for future empirical work; preliminary observations suggest even modestly compositional responses exceed the typical $\mathcal{O}(90)$ -second budgets used in large-scale RLHF programs.

Condition 3.1 (The Hallucination Regime). There exists a plausible fabrication r_{fab} and contexts $(q, w_A), (q, w_B)$ from Condition 2.1 such that:

1. The cost to verify r_{fab} in both contexts exceeds B (e.g., obscure or non-existent citation).
2. Consequently, the supervisor observation is identical:

$$Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B) = (q, r_{fab}, \emptyset)$$

Theorem 3.1. *No learning algorithm $\mathcal{A}$ optimizing reward R from a bounded supervisor S can converge to a Grounded Policy that satisfies Epistemic Honesty across both contexts w_A and w_B for the response r_{fab} .*

Proof. 1. The learning update $\Delta\theta$ is a function of the observation and reward.

2. From Condition 2.2, the observations for r_{fab} are identical in both worlds.

3. Therefore, the reward distribution and expected gradient updates are identical: $\mathbb{E}[\Delta\theta|w_A] = \mathbb{E}[\Delta\theta|w_B]$.

4. The optimizer receives no signal to distinguish the cases.

5. To satisfy Epistemic Honesty, the policy must converge to output r_{fab} in w_A and $\perp$ in w_B .

6. However, since the gradient signal is identical, the policy must converge to the same behavior in both cases (either rewarding r_{fab} in both, or penalizing it in both).

7. It is therefore impossible to learn the split behavior required for honesty. □

Remark 3.1 (Empirical Bias). While the theorem proves only that the behavior must be *identical*, empirical observation shows a strong bias. Supervisors typically prefer “plausible completion” over “abstention” when truth is unknown. This effectively breaks the tie in favor of fabrication.

¹See, e.g., OpenAI, “GPT-4 System Card,” 2023 (Appendix B) and Anthropic, “Constitutional AI: Harmlessness from AI Feedback,” 2022.

Lemma 3.2 (Observation Monotonicity). *Consider any stack of supervisors $(S_1, \dots, S_k)$ operating under the text-only observation model. Let $Obs_i(q, r, w)$ denote the observation exported to S_i . If each supervisor can only act on the exported interface emitted by the previous layer, then there exist deterministic functions f_i such that $Obs_{i+1}(q, r, w) = f_i(Obs_i(q, r, w))$ for all $i < k$. Consequently, later judges see no strictly finer epistemic information than earlier ones, regardless of whether their judgments are deterministic or probabilistic.*

Proof. By definition, each supervisor consumes only the serialized response (and any metadata permitted by the text-only observation model) and produces a finite judgment that forms the entire input to the next layer. Because no layer can query the internal world state directly or issue additional interactive probes, the exported observation available to S_{i+1} must be a deterministic function of Obs_i . Therefore, the informational content across the stack is monotonically non-increasing, and probabilistic scoring cannot introduce distinctions absent from Obs_i . $\square$

Corollary 3.3 (Induction on Layered Judges). *For k -layered supervisors (S_1 supervising π , S_2 supervising S_1 , $\dots$, S_k supervising S_{k-1}):*

- **Base case.** *For $k = 1$, Theorem 3.1 holds (a single bounded supervisor S fails).*
- **Induction step.** *Assume the statement holds for $k = m$. By Lemma 3.2, S_{m+1} receives observations that are deterministic functions of Obs_m , and thus inherits both the verification budget B and the indistinguishability of plausible fabrications. No learning or judging signal distinguishes w_A from w_B , even if S_{m+1} produces probabilistic confidence scores. Therefore the hallucination deadlock persists for $k = m + 1$.*

Artifact: Stacking alignment layers in OLMo-3 (base $\rightarrow$ SFT $\rightarrow$ Instruct) reduces but does not eliminate fabrications ($\sim 50\%$ in the base model to $\sim 30\%$ after alignment, per experimental results).

By induction, no finite k resolves the deadlock.

- **Implication.** *Layering amplifies computational and supervisory costs without fixing the epistemic gaps; an interface redesign that incorporates the world state w at inference time is required. While information-loss non-composability is classical in systems theory, this corollary shows that popular “auditor-of-auditor” proposals inherit the same failure mode so long as they operate under text-only observation models.*

Remark 3.2 (Scope of Layering). The inductive result applies only to verification stacks whose layers observe and transform the same exported response channel. Architectures that surface additional epistemic state between layers—such as retrieval traces, entity resolution tables, or cryptographically bound provenance—leave the text-only observation model and therefore fall outside the impossibility class. Likewise, interactive re-querying that reveals new internal state would constitute a different observation model rather than a “stacked judge” within this theorem.

3.1 The Composition Cost Barrier

The bounded-supervisor result presumes that verification cost grows rapidly with the structural complexity of the model’s response. We capture this intuition with a graph abstraction.

Definition 3.3 (Composition Graph). Given a response r , define its composition graph $G(r) = (V, E)$ where each node $v \in V$ represents an atomic subclaim, named entity, or intermediate entailment, and each edge $(u, v) \in E$ encodes a dependency that must hold for r to remain coherent.

Edges may represent entailment/support relations, co-reference constraints, causal ordering, or temporal consistency requirements.

Lemma 3.4 (Superlinear Verification Cost). *Under a text-only observation model, reconstructing $G(r)$ from the exported string requires the supervisor to infer both the nodes and their dependencies directly from r . Even if parsing each node is constant-time, checking cross-claim consistency requires touching every edge, so the verification cost is $\Omega(|E|)$. For natural-language responses where each subclaim participates in multiple constraints (co-reference, causality, temporal order), $|E|$ grows superlinearly in $|V|$. Consequently, bounded supervisors must either abandon exhaustive checks or resort to sampling once the number of subclaims exceeds a modest threshold.*

Remark 3.3 (Implications for Layering). Corollary 3.3 shows that stacking supervisors without exposing additional epistemic channels merely replicates the same verification bottleneck and often exacerbates it, because each layer must reconstitute $G(r)$ before rendering judgment. Exporting structured traces—for example, explicit subclaim/entity tables with provenance—reduces the reconstruction burden, allowing the bounded budget to be spent on actual verification rather than graph discovery. Industry interfaces that cap human raters at $\mathcal{O}(90)$ seconds per prompt implicitly assume at most 5–7 atomic checks; once responses exceed that composition, raters revert to sampling, which makes plausible fabrications indistinguishable for every layer in the stack.

Corollary 3.5 (Responsibility Concentration). *Under a text-only observation model, epistemic honesty cannot be externally verified by users or downstream auditors. Responsibility for epistemic honesty therefore resides solely with the system owner, who retains access to the internal causal state.*

External attestation mechanisms—cryptographic hash chains, trusted execution environments, or tamper-evident logs—do not contradict this corollary; rather, they reinforce it. Such mechanisms only create verifiable guarantees when the system owner chooses to bind and export epistemic traces, so the authority to make honesty attestable still rests with the owner. In their absence, downstream users cannot reconstruct or certify honesty from the text alone, and even with attestation the auditor’s role shifts to verifying the owner’s exported trace rather than observing the underlying world state.

4 Empirical Diagnostics (Forensic Exhibits)

4.1 The Sycophancy Trap (Exhibit A: Grok)

Artifact: Forced choice between two lies regarding a non-existent Adam Smith paper.

The Failure: The RLHF interface forced the human operator to select a “preferred” lie. This validates **Theorem 2**: The supervisor (human) is bounded and the interface (protocol) cannot express the necessary distinction, forcing a gradient update that reinforces fabrication.

4.2 The Signal-to-Noise Asymmetry (Exhibit B: OLMo)

Artifact: 92% fabrication rate on fictitious papers.

The Failure: Base models produce coherent 1500-token fabrications with negligible “uncertainty” signaling. This validates **Theorem 1**: The predictor-centric base model, lacking access to world-state w , generates the statistically most probable continuation for the query context.

4.3 Architectures Outside the Impossibility Class

Our results do not preclude epistemically honest systems in general. They apply specifically to systems operating under text-only observation models. Architectures that export structured epistemic traces—such as entity resolution candidates, retrieval queries and results, and claim-to-evidence bindings—fall outside this impossibility class.

However, exporting such traces introduces a secondary verification problem: the trace itself must be bound to actual system execution (e.g., via cryptographic attestation or tamper-evident logging). We leave formal treatment of trace verifiability to future work.

5 Architectural Implications

To escape these impossibility results, systems must adopt specific design principles:

Principle 5.1 (State Exteriority). The representation of validity must be separated from the representation of generation. The policy must explicitly condition on retrieved world-state w , not just query q .

Principle 5.2 (Verification Independence). Verification signals must originate from a channel orthogonal to the generation signal, capable of overriding the bounded supervisor’s preference for plausibility.

Principle 5.3 (Provenance Binding). Every output assertion must be structurally bound to a verifiable source.